

How to build a forecasting system?

«Forecasting system architecture»

Forecasting system architecture

Forecasting system architecture components:

- **Component 1 — Data Engineer:** M4 Daily (TSF), ACF, `src/data_loader.py`.
- **Component 2 — ML Engineer:** `src/features.py`, CatBoost- first-differencing MAD- `(src/models/catboost_model.py)`, `(run_experiment.py, src/experiments.py)`.
- **Component 3 — Analyst / Researcher:** `(src/evaluation.py)`, `(src/models/)`, `(analysis_results.ipynb)`.

1. ■■■■■■■■■■ ■■■■■■ ■ ■■■■■■■■■■

1. **Содержание** — текст, который отображается в заголовке документа. Этот текст отображается в заголовке документа, а также в заголовке документа.

შენიშვნა: ზემოთხსენებული მითითებები მოქმედებს მხოლოდ (გადასაყვანილი მითითებები, გადამყვანი მითითებები, მითითებები-გადასაყვანილი) მითითებები მითითებები მითითებები მითითებები CatBoost-მითითებები და მითითებები და მითითებები მითითებები მითითებები მითითებები. მითითებები მითითებები მითითებები მითითებები და მითითებები მითითებები მითითებები და მითითებები მითითებები მითითებები მითითებები.

XXXXXXXXXXXX XXXXXXXXXXXX XXXXXXXXXXXX XX XXXX XXXXXXXXXXXXXXX:

- [illegible]

2. ■■■■■■ ■■■■■■

[illegible]

XXXXXXXXXX XXXXXXXX XXXX XXXXXXXXXXXXXXXXXXXX:

- $\sum_{i=1}^n \sum_{j=1}^n |x_{ij}| \geq 700$ (number of non-zero elements in the matrix).
- $|ACF(\text{lag}=7)| \geq 0.20$ — autocorrelation at lag 7 is significant.

■■■■■ ■■■■■■■■■■ ■ ■■■■■■■ ■■■■■■ 2 745 ■■■■■, ■■ ■■■■■■■ ■■■■■■■■
(random seed=42) ■■■■■■■■ 150 ■■■■■ ■■■ ■■■■■■■■■■■■■■■■■■.

■■■■ ■■■■■■■■■■ ■ ■■■■■■ (|ACF(7)|) ■■■■■■■■■■ ■■ 0.72 ■■ 0.995, ■■■■
 ■■■■■■■■■■ ■■■■■■■■ ■■■■■■ ■■■■■■■■ ■■■■■■■■■■ ■■ ■■■■
 ■■■■■■■■■■ ■■■■■■.

A histogram showing the distribution of seasonality strength, measured as the absolute value of the Autocorrelation Function at lag 7 ($|ACF(lag=7)|$), for 100 time series. The x-axis represents the seasonality strength, ranging from 0.75 to 1.00. The y-axis represents the number of series, ranging from 0 to 100. The distribution is highly right-skewed, with a peak of 100 series at the 1.00 bin.

Seasonality Strength Bin	Number of Series
0.72 - 0.74	1
0.83 - 0.85	1
0.87 - 0.89	1
0.89 - 0.91	1
0.91 - 0.93	1
0.93 - 0.95	2
0.95 - 0.97	4
0.97 - 0.99	14
0.99 - 1.00	21
1.00 - 1.01	100

Last 365 days of 4 sample M4-Daily series (weekly seasonality)



**[REDACTED] [REDACTED] [REDACTED], [REDACTED] [REDACTED] [REDACTED]
[REDACTED].**

- **Naive** — `statsforecast.models.Naive(h=H)`.
- **SeasonalNaive** — `statsforecast.models.SeasonalNaive(h=H, season=7)`.
- **AutoETS** — `statsforecast.models.AutoETS(AIC=statsforecast.AIC)`.
- **AutoTheta** — `statsforecast.models.AutoTheta(theta=Theta, AIC=statsforecast.AIC)`.
`statsforecast.models.Naive(h=H, season=7)`, `statsforecast.models.SeasonalNaive(h=H, season=7)`.

4.2 **Gradient Boosting: CatBoost + 7**

CatBoost (gradient boosting) is a powerful machine learning library. It supports categorical features and is easy to use. It is a good choice for time series forecasting. The `catboost` package is available on PyPI. The `catboost` package is available on PyPI. The `catboost` package is available on PyPI.

Hyperparameters: `iterations=500`, `learning_rate=0.05`, `depth=6`, `loss=RMSE`, `early_stopping_rounds=50`. The `catboost` package is available on PyPI. The `catboost` package is available on PyPI. The `catboost` package is available on PyPI.

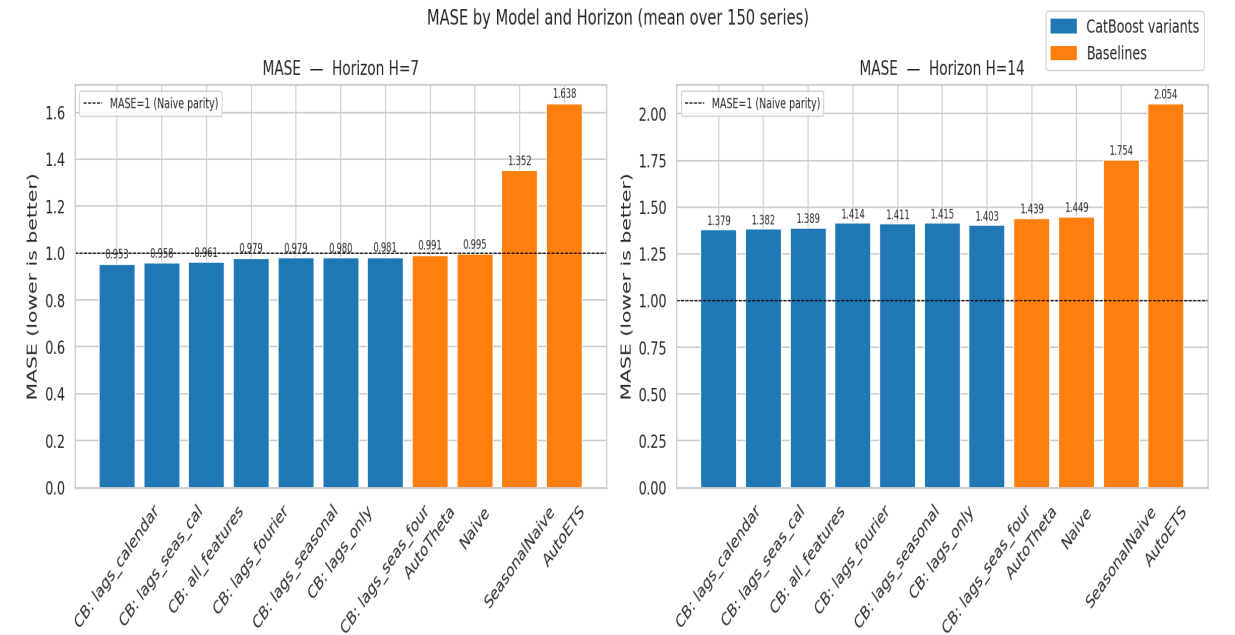
Feature	Feature	Count
lags_only	lags 1-6	6
lags_seasonal	lags 1-6 + lags 7,14,21,28	10
lags_calendar	lags 1-6 + calendar features (month, day, week, quarter, year, ...)	11
lags_fourier	lags 1-6 + fourier features (T=7, K=3; T=365, K=5)	22
lags_seasonal_calendar	lags 1-6 + seasonal features + calendar features	15
lags_seasonal_fourier	lags 1-6 + seasonal features + fourier features	26
all_features	all features	31

5. ■■■■■■■■■■■■

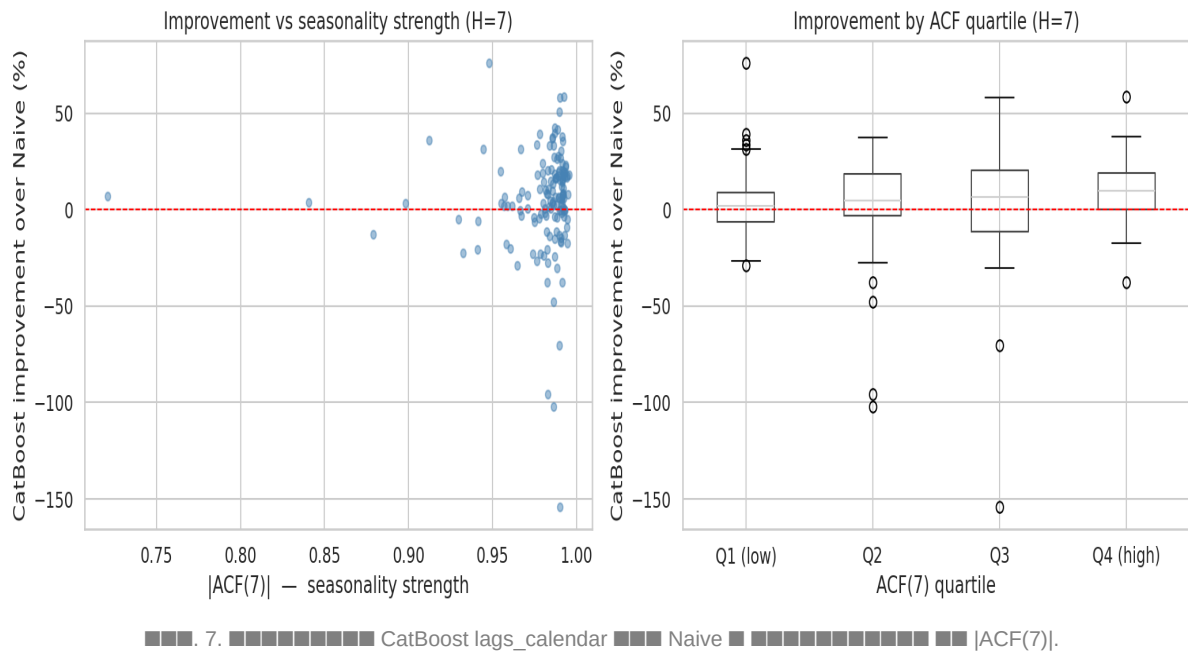
5.1 ■■■■■■■■ ■■■■■■■■ ■■■■■■■■

Model	MAE@H7	MAE@H14	MASE@H7	MASE@H14
CatBoost_lags_calendar	126.2	201.1	0.9532	1.3790
CatBoost_lags_seasonal_calendar	127.5	201.9	0.9584	1.3822
CatBoost_all_features	128.1	203.2	0.9612	1.3889
CatBoost_lags_fourier	131.2	207.7	0.9785	1.4141
CatBoost_lags_seasonal	131.1	206.9	0.9786	1.4108
CatBoost_lags_only	131.4	207.9	0.9798	1.4152
CatBoost_lags_seasonal_fourier	130.8	205.2	0.9814	1.4032
AutoTheta	134.1	212.0	0.9908	1.4392
Naive	134.3	213.6	0.9953	1.4488
SeasonalNaive	191.8	262.1	1.3517	1.7538
AutoETS	177.4	253.9	1.6379	2.0545

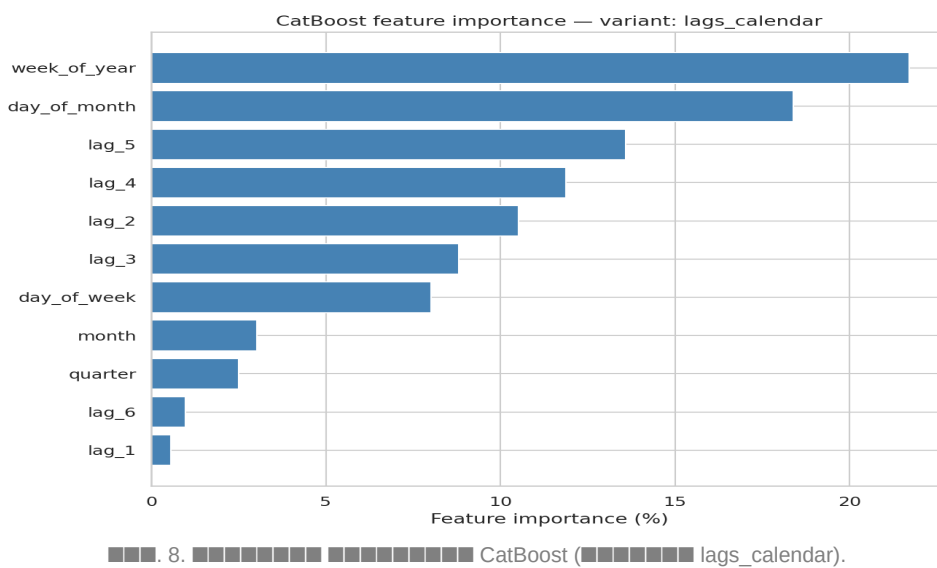
■■■■■■■ 1. ■■■■■■■■ ■■■■■■■■ ■■■■■■■■ 150 ■■■■■■■■ (■■■■■■■■■ ■■■■■■■■ ■■■■■■■■).



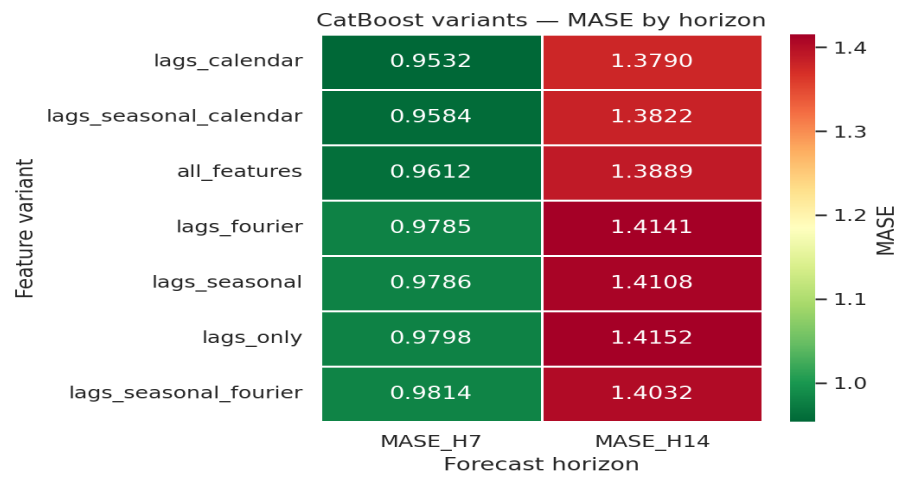
■■■■. 4. MASE ■■■■■■■■ ■■■■■■■■ ■■■■■■■■ H=7 ■ H=14. ■■■■■■■■ = CatBoost-■■■■■■■■■, ■■■■■■■■ = ■■■■■■■■■■.



5.4 **lags_calendar**



5.5 **CatBoost**



9. MASE 7 CatBoost H=7 H=14 (=).

6. ■■■■■■

6.1 ■■■■■■ ■■■■■■

```

##### lags_calendar (##### 1-6 + #####,
##### MASE = 0.9532 ##### H=7 ■ 1.379
##### H=14, ##### Naive (0.995 / 1.449) ■ AutoTheta (0.991 / 1.439) —
#####.

```

6.2 ■■■■■ ■■ ■■■■■■■■■■

- **Model selection: MASE**? **Model selection: MASE**
(**Model selection: MASE** **Model selection: MASE** **Model selection: MASE**) **Model selection: MASE** +2.7% MASE
Model selection: MASE **Model selection: MASE**. **Model selection: MASE** **Model selection: MASE** +0.1% **Model selection: MASE**
Model selection: MASE **Model selection: MASE**. **Model selection: MASE**-**Model selection: MASE** **Model selection: MASE** **Model selection: MASE**
Model selection: MASE **Model selection: MASE** — **Model selection: MASE**, **Model selection: MASE**-**Model selection: MASE** **Model selection: MASE** **Model selection: MASE** lag-7.
Model selection: MASE **Model selection: MASE** **Model selection: MASE** **Model selection: MASE** **Model selection: MASE**
Model selection: MASE **Model selection: MASE**-**Model selection: MASE**.
- **Model selection: MASE**? **Model selection: MASE**: **Model selection: MASE**
Model selection: MASE **Model selection: MASE** (**Model selection: MASE** **Model selection: MASE** **Model selection: MASE** $|ACF7| \geq 0.20$) **Model selection: MASE** CatBoost **Model selection: MASE**
Naive **Model selection: MASE** **Model selection: MASE** **Model selection: MASE** **Model selection: MASE**. **Model selection: MASE** **Model selection: MASE**
Model selection: MASE-**Model selection: MASE** **Model selection: MASE** **Model selection: MASE**, **Model selection: MASE** **Model selection: MASE** **Model selection: MASE**
Model selection: MASE.
- **Model selection: MASE**? MASE CatBoost-**Model selection: MASE** **Model selection: MASE**
Model selection: MASE **Model selection: MASE** **Model selection: MASE** **Model selection: MASE** **Model selection: MASE** **Model selection: MASE** **Model selection: MASE**. **Model selection: MASE**
Model selection: MASE CatBoost **Model selection: MASE** **Model selection: MASE** **Model selection: MASE**.

6.3 ■ ■ ■ ■ ■ ■ ■ ■ ■ ■

- **Naive** — `Naive` (MASE 1.35–1.75): `Naive` (MASE 1.35–1.75): `Naive` (MASE 1.35–1.75):
- **SeasonalNaive** — `SeasonalNaive` (MASE 1.35–1.75): `SeasonalNaive` (MASE 1.35–1.75): `SeasonalNaive` (MASE 1.35–1.75):
- **AutoETS** — `AutoETS` (MASE 1.64–2.05): `AutoETS` (MASE 1.64–2.05): `AutoETS` (MASE 1.64–2.05):
- **AutoTheta** — `AutoTheta` (MASE 0.991–1.000 H=7), `AutoTheta` (MASE 0.991–1.000 H=7), `AutoTheta` (MASE 0.991–1.000 H=7),

6.4 ■■■■■■■■■■ ■■■■■■■■■■

[illegible]

- 1. `first-differencing` `lags_calendar`` `all_features``.
- 2. `lags_calendar`` `all_features`` `lags_calendar`` `all_features`` (`lags_calendar`` `all_features``) `lags_calendar`` `all_features`` `lags_calendar`` `all_features``.
- 3. `lags_calendar`` `all_features`` `lags_calendar`` `all_features``: `lags_calendar`` `all_features``.
- 4. AutoTheta — `lags_calendar`` `all_features``, `lags_calendar`` `all_features`` `lags_calendar`` `all_features`` (`lags_calendar`` `all_features``).

■■■■■ ■■■■■